

Ankan Bhattacharyya

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EDUCATION

University of Kentucky

Lexington, KY

PhD in Computer Science; GPA: 3.8/4.0

Aug 2021 - Aug 2026

Courses: Computer Graphics, Computer Vision, Biomedical Imaging, NLP, Algorithms, Machine Learning, Linear Algebra

Thesis: Non-invasive digital restoration of damaged photographic film negatives using X-ray and machine learning (Supervisor: Brent Seales, PhD)

West Bengal University of Technology

West Bengal, India

Bachelor of Technology in Computer Science and Engineering; GPA: 8.59/10.0

Aug 2013 - May 2017

Thesis: Recognition of Online Handwritten Bangla Words (Supervisor: Shibaprasad Sen, PhD)

SKILLS

Technical: Artificial Intelligence, Machine Learning, Pattern Recognition, Deep Learning, Computer Vision, Transformers, Image Reconstructions, Structure from Motion, Multi View Stereo, Photogrammetry, Photographic Film Negative development

Languages: Java, C++, Python, C, SQL, Unix scripting

Libraries and Framework: OpenGL, OpenCV, OpenMVG, OpenMVS, PyTorch

Tools and Services: Git, Docker, Singularity, AWS, Autodesk Fusion360, Adobe Lightroom, FIJI, Adobe Premiere Pro, XMT scan(SKYSCAN 1273), 3D Printing, 3D Printer (Bambu Labs)

EXPERIENCE

Teaching experience

Graduate Teaching Assistant

August 2025 - December 2025

University of Kentucky

Lexington, KY

- Served as a grader for Algorithm and data structure
- Wrote unit tests for grading coding assignments

Graduate Teaching Assistant

January 2024 - May 2024

University of Kentucky

Lexington, KY

- Served as primary instructor teaching Undergraduate 2D Graphics and Image Processing using JAVA (CS335)
- Designed syllabus and examinations for the course
- As part of assignments, the students were required to code and complete fun games like the minesweeper using flood fill algorithm, hoops using Bézier curve and Splines, image morphing, card matching game and image warping
- Discussed the basics of camera system, including pin hole camera and how images are understood
- Discussed rendering and double buffering
- Discussed linear algebra required to solve image translation problems in 2D graphics
- Discussed how rasterization worked
- Provided Letter of Recommendation to student

Graduate Teaching Assistant

August 2021 - May 2022

University of Kentucky

Lexington, KY

- Served as lab instructor and grader teaching Undergraduate 2D Graphics and Image Processing using JAVA (CS335)
- Conducted lab for computer graphics
- Graded exams and held office hours for doubt clearing
- Taught classes on image processing

Research experience

Graduate Student (Thesis)

University of Kentucky

Principal Investigator: Brent Seales, PhD & Seth Parker, PhD

May 2024 - August 2026

Lexington, KY

- Worked on non invasive digital restoration of damaged photographic films
- Used X-ray Computed micro-Tomography to scan film rolls
- Developed automated segmentation for segmenting the films and then opening it virtually
- Developed machine learning pipeline to generate print quality images from X-ray rendering of photographic films
- Studied film development and physical restoration of damaged photographic film negatives
- Developed imaging system pipeline to generate high dynamic range (HDR) image from film negatives

Graduate Research Assistant

EduceLab & Jim Beam

Principal Investigator: Brent Seales, PhD & Seth Parker, PhD

August 2023 - May 2024

Lexington, KY

- Developed deployable camera system to image rolling barrels in production line of Jim Beam
- Used photogrammetry to generate multi-view structure of barrels and virtually flatten the barrel structure
- Developed AI pipeline to segment the barrel staves and hoops
- The work is presented at AI in Practice Seminar
- Scanned scroll fragments from ship-wreck and Jordan
- Worked and guided undergraduate and high school students regarding their projects, provided with scans and helped develop pipeline to render the hidden materials from the scans

Graduate Research Assistant

EduceLab

Principal Investigator: Brent Seales, PhD & Seth Parker, PhD

August 2022 - August 2023

Lexington, KY

- Worked on *Smithsonian 3D Viewer* to view Herculaneum Scroll fragment on web
- Developed 3D rendering pipeline viewer for viewing Herculaneum Scroll fragment from the data acquired by photogrammetry

Graduate Research Assistant

EduceLab

Principal Investigator: Brent Seales, PhD & Seth Parker, PhD

June 2022 - August 2022

Lexington, KY

- Developed multi-spectral camera system to image damaged sacramental scroll preserved from *Holy Trinity Greek Orthodox Church, New Orleans* .
- Developed an automated pipeline to recover washed away inks from the sacramental journal
- The work is published here and presented at KCS seminar

Independent Researcher

Jadavpur University

Principal Investigator: Shibaprasad Sen, PhD & Ram Sarkar, PhD & Kaushik Roy, PhD

January 2014 - August 2021

Kolkata, West Bengal, India

- Worked on bangla online handwriting recognition
- Developed AI pipeline for document analysis
- Developed groundtruth tool for online handwritten bangla document
- Published all the works at reputed international journals, like Scientific Reports (Nature) and ACM TALLIP and international conferences

Industrial experiences

AWS Developer

Cognizant Technology Solutions

August 2017 - July 2021

Bantala, West Bengal, India

- Developed and deployed new notification framework, that helped emailing and sms opt-in during hotel booking, and implemented in layers, that is used by several different projects being developed by the client
- Developed pipeline to store data to AWS Redshift (datawarehouse) for Dunkin Brands Integration
- Developed a voice enabled, multi-lingual search engine, that helped reduce the number of clicks drastically to a single click, thus increasing hotel booking by a decent amount
- Implemented GDPR for Wyndham Hotels and Resorts

PUBLICATIONS

- **Ankan Bhattacharyya**, "Non-invasive digital restoration of damaged photographic film negatives", PhD Thesis, University of Kentucky (2026) (<https://doi.org/10.13023/etd.2026.411>)
- **Ankan Bhattacharyya**, C Seth Parker, W Brent Seales, "Multispectral Imaging of Damaged Sacramental Journal Pages: A Preliminary Study", Proceedings of 4th International Conference on Frontiers in Computing and Systems. COMSYS 2023. Lecture Notes in Networks and Systems, vol 975. Springer, Singapore. (2024) (<https://doi.org/10.1201/9781003205326>)
- **Ankan Bhattacharyya**, Soumyajit Saha, Shibaprasad Sen, Seyedali Mirjalili, Ram Sarkar, "Deep Feature Selection Using Moth-Flame Optimization for Facial Expression Recognition from Thermal Images", Handbook of Moth-Flame Optimization Algorithm: Variants, Hybrids, Improvements, and Applications (1st ed.). CRC Press. (2022) (<https://doi.org/10.1201/9781003205326>)
- **Ankan Bhattacharyya**, Rajatshubra Chakraborty, Soumyajit Saha, Shibaprasad Sen, Ram Sarkar, Kaushik Roy, "A Two-Stage Deep Feature Selection Method for Online Handwritten Bangla and Devanagari Basic Character Recognition", SN Computer Science, Progresses in Image Processing (2022) (<https://doi.org/10.1007/s42979-022-01157-2>)
- **Ankan Bhattacharyya**, Somnath Chatterjee, Shibaprasad Sen, Aleksandr Sinitca, Dmitrii Kaplun, Ram Sarkar, "A deep learning model for classifying human facial expressions from infrared thermal images", Scientific Reports, Nature (2021) (<https://doi.org/10.1038/s41598-021-99998-z>)
- Shibaprasad Sen, **Ankan Bhattacharyya**, Ram Sarkar, Kaushik Roy, "BYANJON: A Ground Truth Preparation System for Online Handwritten Bangla Documents", ACM Transactions on Asian and Low-Resource Language Information Processing (2021) (<https://doi.org/10.1145/3464379>)
- Rajatshubra Chakraborty, Soumyajit Saha, **Ankan Bhattacharyya**, Shibaprasad, Sen, Ram Sarkar, K. Roy, "Recognition of Online Handwritten Bangla and Devanagari Basic Characters: A Transfer Learning Approach", 5th IAPR International Conference on Computer Vision & Image Processing (IAPR CVIP2020) (2020) (https://doi.org/10.1007/978-981-16-1092-9_45)
- Rajatshubra Chakraborty, Debadrita Mukherjee, **Ankan Bhattacharyya**, Himadri Mukherjee, Monoj Kumar Sur, Shibaprasad Sen, Kaushik Roy, "Online Handwritten Bangla and Devanagari Character Recognition by using CNN: A Deep Learning Concept", IEEE 1st International Conference for Convergence in Engineering (ICCE) (2020) (<https://doi.org/10.1109/ICCE50343.2020.9290566>)
- Shibaprasad Sen, Mridul Mitra, **Ankan Bhattacharyya**, Ram Sarkar, Friedhelm Schwenker, Kaushik Roy, "Feature Selection for Recognition of Online Handwritten Bangla Characters", Neural Processing Letters (2020) (<https://doi.org/10.1007/s11063-019-10010-2>)
- Shibaprasad Sen, **Ankan Bhattacharyya**, Kaushik Roy, "The Effect of Using Features Computed from Generated Offline Images for Online Bangla Handwritten Character Recognition", Document Processing Using Machine Learning (2019) (<https://doi.org/10.1201/9780429277573>)
- Shibaprasad Sen, **Ankan Bhattacharyya**, Mridul Mitra, Kaushik Roy, Sudip Kumar Naskar, Ram Sarkar, "Online Bangla handwritten word recognition using HMM and language model", Neural Computing and Applications (2019) (<https://doi.org/10.1007/s00521-019-04518-w>)
- Shibaprasad Sen, **Ankan Bhattacharyya**, Pawan Kumar Singh, Ram Sarkar, Kaushik Roy, David Doermann, "Application of structural and topological features to recognize online handwritten Bangla characters", ACM Transactions on Asian and Low-Resource Language Information Processing (2018) (<https://doi.org/10.1145/3178457>)
- Shibaprasad Sen, **Ankan Bhattacharyya**, Avik Das, Ram Sarkar, Kaushik Roy, "Design of novel feature vector for recognition of online handwritten Bangla basic characters", Proceedings of the First International Conference on Intelligent Computing and Communication (2017) (https://doi.org/10.1007/978-981-10-2035-3_50)

REVIEWER

- Journal: ACM Transactions on Asian and Low-Resource Language Information Processing (ACM TALLIP), SN Computer Science
- Conferences: Official Reviewer for International Conference on Pattern Recognition, 2024 (ICPR-2024), 3rd International Conference on Data, Electronics and Computing, 2024 (ICDEC-2024), Pervasive and Mobile Computing

HONORS AND AWARDS

- Outstanding Student Paper Award, "A deep learning model for classifying human facial expressions from infrared thermal images" University of Kentucky, Department of Computer Science, April 2022.
- UniPi Merit Scholarship for Master's Degree (Laurea Magistrale) in Computer Science A.Y. 2021/2022 awarded by University of Pisa, Italy (Did not avail)
- Participated in "Young IT Professional Award" (YITPA) organized by Computer Society of India, in 2020.
- Most Valuable Player (MVP) 2019, from Travel & Hospitality, Cognizant for outstanding performance in contributing to the Domain.
- First prize in Ideathon 2018 Hackathon in Cognizant, as a team.
- The Best Project Award by Cognizant, in 2017.